

SCIENTIFIC RESEARCH SUPPORT SERVICES

Science made simple. Data made meaningful.

Specialist research, analysis and communication support for organisations that run on science — without the cost of another full-time hire.

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INTRODUCTION

Keerthana Gunaretnam

Computational biologist and independent research consultant — biology, bioinformatics and space science under one roof.



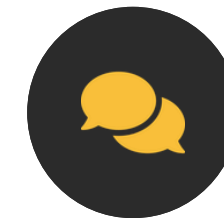
Background

Computational biology and disease-omics, with a second specialism in astrobiology and planetary science. Interdisciplinary researcher at araCreate Group and Computational Biologist (Bio-Intelligence Systems) at Armortrix Technologies, a deep-tech company in Chennai.



What I am good at

Turning raw scientific data into analysis, figures and writing a board, a reviewer or a customer can actually use — sequence analysis, statistics, literature work and science communication.



How I work

Directly with you, under a written scope and either a fixed price or a monthly plan. No account managers and no handovers — the person who runs the analysis also writes the explanation.

THE PROBLEM

Science is expensive to hire — and easy to waste

Three bottlenecks show up again and again in science-driven organisations.



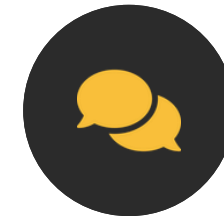
Hiring is the wrong size

Most projects need six weeks of a scientist, not a permanent salary, benefits and onboarding. So the work waits — or it goes to someone who isn't trained for it.



Data sits unused

Sequencing runs, assay results, trial data and field surveys pile up on drives. Value only appears once someone cleans, analyses and interprets them properly.



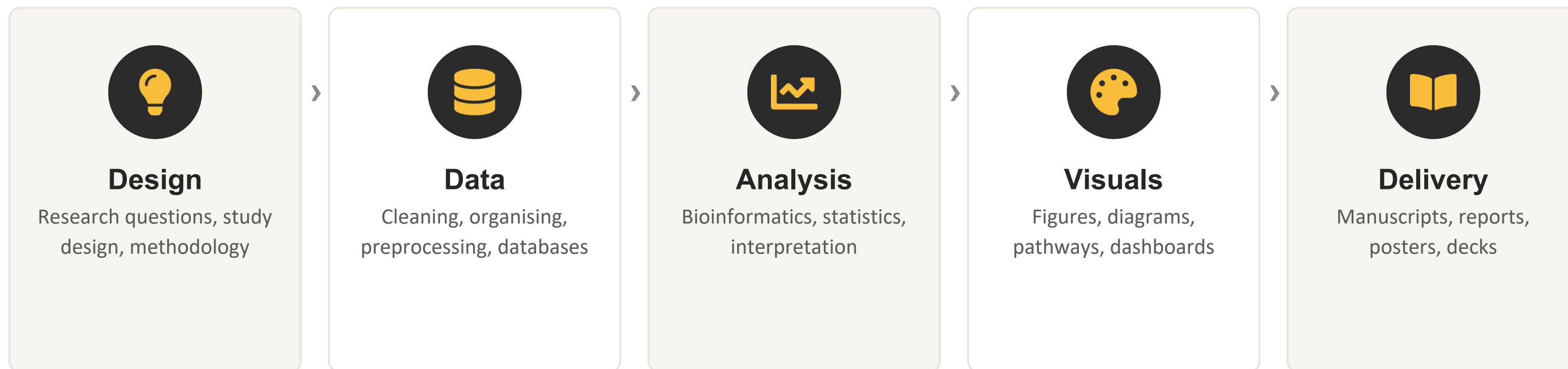
Findings that don't land

Grants, deals and papers are lost on presentation. Strong findings buried in a messy figure or an unreadable report simply don't land with reviewers, boards or buyers.

THE OFFER

One specialist across the whole research chain

Come in at any stage. Most clients start with one step and hand over more once the work proves itself.



The point of difference: the same person who runs the analysis also builds the figure and writes the explanation — so nothing is lost in translation.

What I do



Biology & biotechnology

Molecular biology, genetics and genomics, microbiology, cell biology and biochemistry, human physiology, project and methodology guidance.



Bioinformatics & data

Sequence analysis, BLAST and database searches, comparative genomics, dataset cleaning, pipelines and workflow support, network analysis.



Research & academic support

Literature reviews and searches, proposals, manuscript and thesis support, editing, referencing, abstracts, report preparation.



Data analysis & visualisation

Statistics, tables and graphs, publication-ready figures, pathway diagrams, graphical abstracts, research infographics.



Presentation & design

Conference and research posters, slide decks, thesis-defence presentations, scientific infographics and content visualisation.



Science communication

Article editing, science blogs and educational content, research summaries, technical content rewritten for non-specialists, LinkedIn content.

Also available

Specialist and long-tail work that clients regularly bundle into a retainer.



Astronomy & planetary science

- Exoplanets and astrobiology
- Space-mission data interpretation
- Astronomical dataset analysis
- Scientific writing on space topics



Training & educational support

- Team upskilling in bioinformatics
- Biology, genetics and molecular biology tutoring
- Research and writing skills coaching
- Presentation and viva preparation



Documents & profiles

- Academic CV and researcher profiles
- Conference abstract formatting
- Poster-to-presentation conversion
- Canva, Excel and PowerPoint builds

Who this is built for

Six kinds of organisation where this support pays for itself fastest.



Biotech & life-science startups

Lean teams that need scientific depth before they can afford a research hire.



Pharma, diagnostics & CROs

Overflow analysis, documentation and figure preparation under deadline.



Agri, food & environment

Field and lab data turned into decisions, reports and regulatory evidence.



Universities & research groups

Proposal, manuscript and thesis support when supervision is stretched thin.



EdTech & scientific publishers

Accurate, well-written biology and biotechnology content at volume.



Space & data organisations

Astronomy and planetary datasets analysed and explained for wider audiences.

DELIVERABLES

What lands in your inbox

Every engagement ends in files your team can use immediately — not a summary of effort.

✓ Cleaned, documented datasets with a reproducible script

✓ Analysis report with methods, results and plain-English interpretation

✓ Publication-ready figures in vector and raster formats

✓ Annotated pathway, mechanism and workflow diagrams

✓ Manuscript, proposal or thesis drafts, fully referenced

✓ Conference posters and slide decks, print-ready

✓ Literature reviews with a screened, citable source list

✓ Editable source files — you keep and reuse everything

Delivered in the format you work in: Word, Excel, PowerPoint, PDF, R/Python scripts, Canva or Google Slides.

PROCESS

Four steps, no surprises

From first message to delivered work, with a fixed scope and price agreed before anything starts.

01

You share the requirement

A brief message or call. Send the data, the paper, the deadline — whatever exists.

02

We agree scope and price

You get a written scope, timeline and fixed quote. Nothing begins until you approve it.

03

I deliver on time

Progress checkpoint at the halfway mark so direction is corrected early, not at the end.

04

You get a result you can use

Delivery plus one round of revisions included, and a walkthrough call if you want one.

Confidentiality as standard: happy to work under your NDA, and your data never appears in any portfolio without written permission.

COMMERCIALS

Three ways to work together

Start with one project, or put me on a monthly plan. Move between them at any time.

FOR ONE-OFF BRIEFS

Per project

Fixed quote per brief

- A figure to redraw, a section to edit
- Quoted after a free scoping call
- One revision round included
- 50% up front, 50% on delivery

MOST POPULAR

Monthly retainer

From \$75 / month

- Reserved capacity every month
- Priority turnaround on requests
- Lower effective rate than ad-hoc
- Rolling, cancel with 30 days notice

FOR ONGOING AWARENESS

Subscription

From \$150 / month

- A written intelligence brief each month
- Papers, funding, patents, competitors
- No scoping conversation each time
- Cancel any time

Rates in USD; LKR equivalent available. Student, academic and non-profit rates on request.

RETAINER PLANS

Monthly research support

Ongoing research capacity for teams that need a scientist regularly, but nowhere near full-time.

FOR OCCASIONAL NEEDS

Starter

\$75 / month

- 2 literature searches
- 2 research summaries
- Reference organisation and formatting
- 1 small research task

MOST POPULAR

Research

\$150 / month

- Weekly literature monitoring
- Research summaries as you need them
- Competitor and landscape scans
- Data organisation
- Scientific writing and editing

FOR CONTINUOUS WORKLOAD

Research Partner

\$250 – 400 / month

- Everything in Research, plus:
- Data analysis and full reports
- Presentations and visualisation
- Regular review meetings

Unused allowance rolls over one month. Move up or down a tier whenever your workload changes.

SPECIALIST PLANS

Two monthly plans for recurring needs

For teams whose repeating need is content or data, rather than general research.

Science content retainer

\$100 – 200 / month

- ✓ 4 LinkedIn posts written in your voice
- ✓ 2 science or technology articles
- ✓ 4 research and news summaries
- ✓ Competitor updates and technical editing

Bioinformatics support

From \$200 / month

- ✓ Sequence analysis, BLAST and database searches
- ✓ Dataset cleaning and genomics analysis
- ✓ Python and R analysis, plus visualisation
- ✓ Send small tasks any time, with no re-quoting

Both plans are rolling. You send work as it comes up, instead of raising a new quote every single time.

SUBSCRIPTION

The Monthly Intelligence Brief

A written brief, delivered as a PDF every month, on what actually moved in your field.



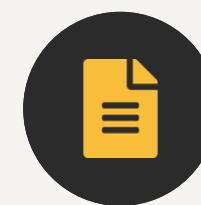
What is inside

- 15 important new papers
- 10 emerging companies
- 5 funding announcements
- 5 competing technologies
- Notable patents and developments



Who it is for

- Biotech and healthtech startups
- Investors and analysts
- Consultants and advisory firms
- Research organisations



How it works

- Delivered on a fixed date each month
- Scoped to your therapeutic or technical area
- A short note on why each item matters
- From \$150 / month, cancel any time

Why teams move from projects to a retainer



Capacity that is already yours

Your work has reserved hours every month, so nothing queues behind another client and nothing waits on a quote.



One scoping call, not twelve

Agree the plan once. After that you send work as it comes up — a search, a figure, a summary — with no admin each time.



A lower effective rate

Monthly plans are priced below ad-hoc work, so the same volume of work costs you less than buying it piece by piece.



Context that compounds

I already know your pipeline, your competitors and your file conventions, so month three is faster and sharper than month one.



A predictable line on the budget

One fixed monthly figure instead of unpredictable invoices — simple to approve internally and simple to defend later.



No lock-in

Rolling month to month, cancel with 30 days notice, and change tier whenever the workload changes.

Why clients keep coming back



Genuinely interdisciplinary

Biology, bioinformatics and astronomy under one roof — rare in a single consultant, and the reason cross-domain briefs work here.



Analysis and communication together

The person who runs the numbers also builds the figure and writes the explanation, so the story survives the handover.



Deadlines treated as fixed

Scope agreed in writing, checkpoint at the halfway mark, delivery on the date promised.



Confidential and professional

Comfortable under NDA. Unpublished data, IP and client identities stay private by default.



Direct, no account managers

You talk to the person doing the work. Questions get answered the same day, not routed through a queue.



Detail that survives review

Methods, references and figures built to hold up in front of reviewers, regulators and technical buyers.

Tools and platforms I work in

Standard, auditable tooling — so your team can open, check and continue the work.



Analysis & scripting

R · Python · Galaxy · Excel



Sequence & structure

BLAST · MAFFT · Clustal Omega · MEGA · Chimera etc



Databases & networks

UniProt · Pfam · InterPro · Cytoscape etc



Figures & delivery

Canva · PowerPoint · Google Slides · Word

Working with a tool that isn't listed? Tell me — most scientific software is a short learning curve away.

EXAMPLE ENGAGEMENTS

What a typical brief looks like

Typical briefs, anonymised — the shape of work most clients start with.

BIOTECH STARTUP

From raw sequences to an investor-ready story

- Brief: comparative genomics on 40 isolates, for a funding round
- Work: cleaning, BLAST and alignment pipeline, annotated figures
- Output: analysis report plus five founder-ready slides

RESEARCH GROUP

A stalled manuscript, submitted in six weeks

- Brief: three years of results, no time to write it up
- Work: literature review, structure, figure rebuild, referencing, full edit
- Output: submission-ready manuscript and cover letter

EDTECH / PUBLISHER

Technical accuracy at content volume

- Brief: biotechnology explainers for a non-specialist audience
- Work: scripting, fact-checking, diagram design, plain-language rewriting
- Output: monthly explainer modules on retainer

GETTING STARTED

An easy, low-risk way to begin

You do not have to commit to a monthly plan on day one.



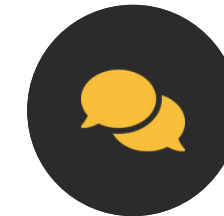
01 A free scoping call

Twenty minutes. Tell me what is sitting on your desk right now. You leave with a written scope and a fixed price, whether or not you go ahead.



02 One small paid task

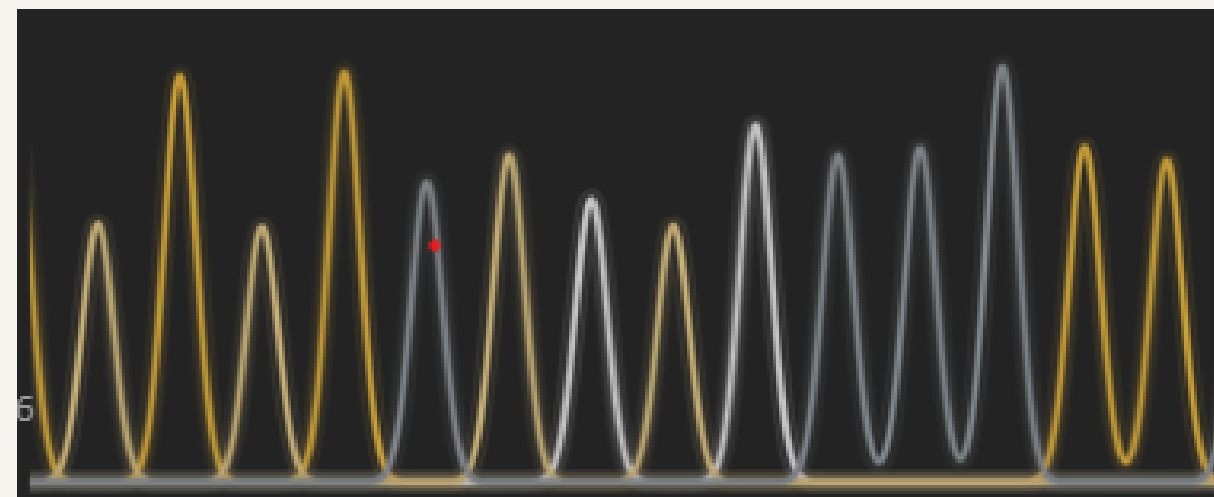
Start with a single literature search, figure or summary. Small enough to be an easy yes, large enough to show you exactly how I work.



03 Pick a monthly plan

If it earns its place, we pick the tier that matches your workload. Most clients begin on Starter or Research and move up from there.

Where impossible ideas become to possible



SangerScope - Sri Lanka

Clinical-grade interpretation for Sanger sequencing.

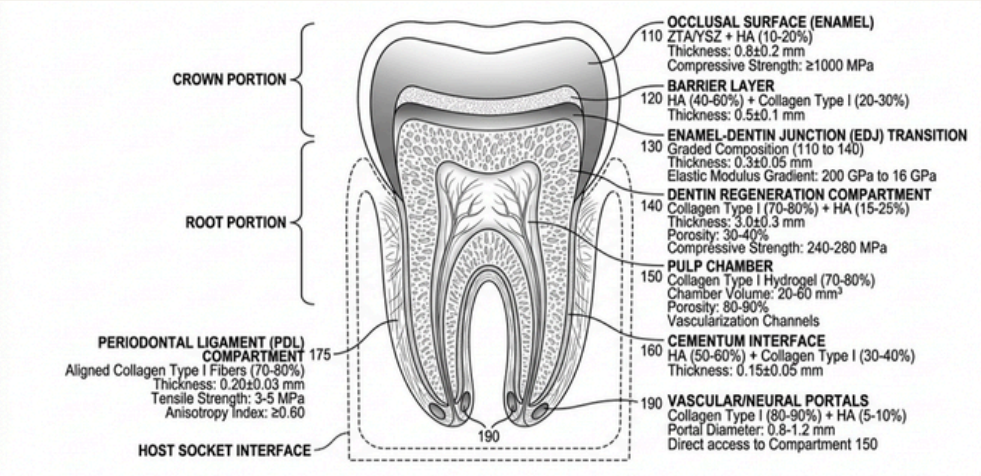
Abstract

Post-market surveillance (PMS) safeguards patients and informs regulatory action by monitoring device performance in real-world use. However, traditional PMS—reliant on manual review, voluntary reports and heterogeneous data—struggles to deliver timely, actionable insights. Artificial intelligence (AI) can modernise PMS through automated triage and signal detection, natural language processing (NLP) for unstructured narratives, and predictive analytics focused on well-labelled outcomes. Building on the journal’s guidance for concise, impartial focus articles, we set out a pragmatic workflow that: (i) employs MAUDE for text-driven signal detection rather than rate estimation; (ii) uses the FDA Medical Device Recalls database for supervised modelling tasks where labels (e.g., recall class, date) are reliable; and (iii) embeds human oversight and transparent documentation. We then add a compact case study leveraging a 20-year analysis of arthroscopic-equipment recalls to show how severity classification and time-to-recall modelling can prioritise CAPA, supplier audits and procurement decisions. This approach aligns with recent commentary emphasising narrow, task-specific AI, structured data and incremental adoption. Taken together, these steps allow manufacturers to strengthen patient safety and regulatory compliance without waiting for fully automated PMS systems.

Keywords: Post Market Surveillance, Artificial Intelligence, Patient safety, Adverse event detection, Regulatory oversight, R program language, MAUDE

MedTech - USA

Post-market surveillance to improve patient safety using R



StartUp - Austria

Biotooth Tissue Engineering

LET'S WORK TOGETHER

Start with one project.

A 20-minute call is enough to scope the first piece of work and put a fixed price on it. No obligation, no charge for the conversation.

- ✓ Send the brief, the data or the draft — whatever exists today
- ✓ Free scoping call and written quote within 48 hours
- ✓ Work starts as soon as you approve the scope

Science is better when shared — let's make it impactful.

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